

Walkbotr: Extracting Walk Bouts from GPS and Accelerometry Data for Physical Activity Research

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Summary

Walking is the most common form of physical activity and a behavior of key interest for urban planners, health promotion researchers, and rehabilitation medicine practitioners. Personal monitoring with accelerometer and Global Positioning System (GPS) devices is the gold standard measurement approach for walking. However, processing accelerometer and GPS device traces to identify walking is a computational and algorithmic challenge. Walkbotr is a new package that allows researchers to process raw accelerometry and GPS traces into standardized walk bouts that can be used to analyze the relationships between urban design, physical activity, and health outcomes. In addition to providing a consistent metric for physical activity analysis, the package de-identifies the original data in a summarized dataset, allowing research groups to use both a full set of data with all identifying information, as well as collaborate on analyses without the complications of working with individually identifiable datasets.

Contribution of walkbotr to Demography and Public Health

Data collected from monitoring devices, such as GPS trackers and accelerometers, hold considerable public health research potential [Feng:2013, Troped:2008]. By analyzing patterns in individual energy expenditure and movement, these data can be used to objectively measure walking and its effects, unlocking a researcher's ability to identify encouragement for, and barriers to, this key cardioprotective behavior across geospatial contexts and populations [Kang:2013, Jankowska:2015].

In their raw form – a series of timestamps, locations, and accelerometer counts – monitoring data are rarely of direct researcher interest, and may also (in the case of GPS data) be identifying, but it can be challenging to process these data efficiently to identify behaviors of interest (e.g. periods of walking) that may be safely shared across research teams. Some methods to identify travel using personal monitoring data (e.g. PALMS, and its successor HABITUS) (<https://www.habitus.eu>) process data on secure servers [Carlson:2015]; however, researchers whose privacy agreements with study participants preclude storing data on 3rd party servers may prefer a local R package. To address this gap, we developed a package, walkbotr, that implements a previously validated algorithm to extract patterns in monitoring data consistent with walking [Kang:2013]. walkbotr allows researchers to (1) process raw personal time

stamp-linked GPS and accelerometry data for identifying periods and locations of walking and (2) create a deidentified summary of walking behavior that can be used in research and practice.

A walk bout is defined as a period of activity in which an accelerometer trace indicates movement consistent with walking and GPS traces from the corresponding time period indicate movement through space consistent with walking (e.g., based on speed) as well.

The inputs of the `walkboudr` package are individual-level accelerometry and GPS data. The output of the package is a data frame integrating all walk bouts (with corresponding times, duration, and summary statistics) identified in those data. A secondary output is a smaller dataset in which all identifying information have been omitted, where this bout summary dataset simply contains a list of all bouts for an individual as well as their corresponding category, median speed, and limited other relevant details. Researchers can use either the full dataset that contains bout labels and categories, or the bout summary with all personally identifying information removed.

By offering a comprehensive set of functions, `walkboudr` empowers researchers and public health practitioners to explore, evaluate, and interpret walk bout data with or without identifying information. This paper introduces the design, features, and potential applications of our R package, highlighting its role in advancing public health research and fostering a deeper understanding of the relationship between physical activity and overall well-being.

Walk-through of `walkboudr`

Definition of a walk bout

The idea of a walk bout derives from the physical activity literature, in which monitored time is partitioned into ‘activity bouts’ and inactive time. **A walk bout is a physical activity bout in which both activity count range and GPS trace is consistent with walking** [Kang:2013].

To identify physical activity bouts, we first classify each epoch (see below for a definition of an epoch) as *active* or *inactive* following Troiano et al [Troiano:2008]. Epochs are defined as active if the accelerometer records more than 500 counts per epoch (CPE) when epochs were set at 30 seconds long and inactive otherwise (values that are parameterized in the package and can be set by the user). This relatively low threshold compared with other physical activity research was selected to allow for capture of slow walking.

An epoch is technically defined as a discrete time interval at which accelerometers collect data – typically accelerometers collect continuous streams of data and divide them into non-overlapping time windows that are referred to as epochs [Troiano:2008]. Within an epoch, the data from the accelerometer is summarized to represent the activity level during that specific time interval [Troiano:2008].

Similarly, CPE refers to the total number of counts that were recorded by the accelerometer within a single epoch duration [Troiano:2008]. CPE serve as a fundamental measure of an individual’s physical activity level over short time intervals. To calculate CPE for a particular epoch, the raw acceleration data for each axis is usually processed by applying filters or mathematical algorithms to remove noise and gravitational effects. Then, the absolute values of the filtered acceleration readings are summed across all three axes to obtain the total count value for that epoch.

Next, a physical activity bout is any contiguous set of epochs that:

- Contains at least 10 cumulative 30-second epochs of being active
- Begins with an active epoch preceded by at least 4 consecutive 30-second epochs of inactivity (i.e., is not part of a prior active epoch)

- Ends with an active epoch followed by at least 4 consecutive 30-second epochs of inactivity

We can then classify physical activity bouts as walking or not walking based on GPS traces. A physical activity bout is a walking bout if it:

- is not a *dwelling bout* (i.e., a bout in which an individual did not leave a pre-specified radius, determined using the GPS trace (default: 66 feet) [Kang:2013], indicating that the individual was not moving in space).
- has a median speed consistent with that of walking (specific thresholds configurable and outlined below)
- has sufficient GPS coverage

How walkbouter works

I. Accelerometry Process

First, from accelerometry data alone, we identify physical activity bouts. These bouts indicate periods of time in which the wearer appears to be physically active, though not necessarily walking (e.g., they could be playing a sport or working out). In order to identify physical activity bouts in accordance with the definition given above, walkbouter uses a run-length encoding algorithm to identify subsequences within the accelerometry data where there are 4 or more consecutive epochs where the activity level is above the threshold indicating the individual was active (>500 CPE). Where 4 or more epochs are inactive, the last of those consecutive epochs is by definition not part of a bout (Figure 1).

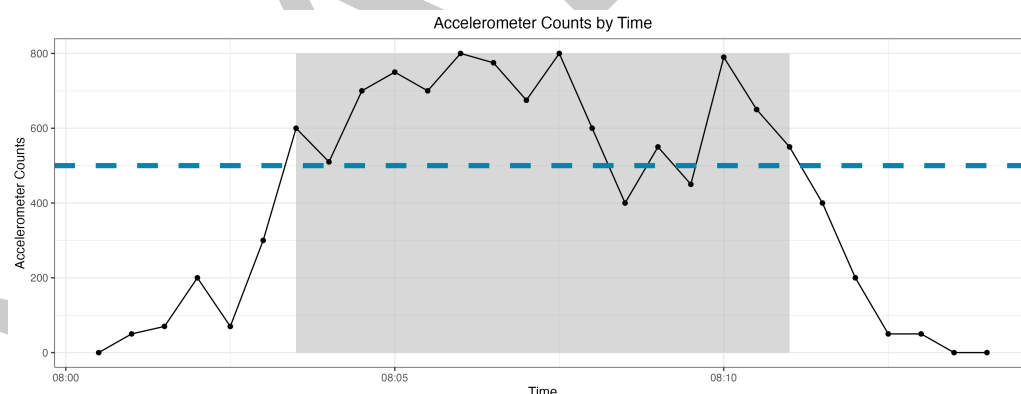


Figure 1: An accelerometry trace indicating a bout, where the bout period is indicated by the gray bounds and the epochs outside of these bounds are not included in the bout. The blue line shows the threshold for an individual being considered active, which defaults to 500 CPE in walkbouter.

The dataset can then be divided into non-bouts and potential bouts (i.e., subsequences in which a beginning and end of a period need to be identified to determine whether the activity periods extend long enough to be an activity bout). The potential bouts can then be run-length encoded to identify bouts. Each sequence of epochs that have been identified as potential bouts will have a number of inactive epochs at the end of the series that is equal to the maximum number of consecutive inactive epochs in a bout, a parameter that can be specified by the user but defaults to 3 epochs (or 1.5 minutes for 30s epochs).

All potential bouts that were found in this first step are now filtered to only include those that have enough active epochs to be considered a bout, another parameter that can be specified by the user but defaults to 10 active epochs (or 5 minutes).

118 If there are no bouts, walkbotr stops and returns a message indicating that there are no
119 bouts. If there are physical activity bouts, walkbotr labels each bout with a numeric value.
120 These bouts will later be labeled as either walk bouts or non-walk bouts.

121 In addition to finding all bouts in the accelerometry data, walkbotr identifies non-wearing
122 periods in the accelerometry data based on a threshold of consecutive epochs with activity
123 counts of 0 CPE. When there are 20 consecutive minutes in which activity counts are 0 CPE,
124 the individual is determined not to be wearing their accelerometry device [Saelens:2014].
125 These periods are flagged and labeled as non-wearing periods.

126 Finally, walkbotr determines if the user wore their accelerometer for a sufficient sampling
127 period of 8-hours, also referred as a "complete day." This 8-hour threshold comes preset as
128 a modifiable constant within the walkbotr package [Saelens:2014]. walkbotr does this
129 by converting time to the local time zone indicated by the user, identifying each full day
130 sequence, and determining if the non-wearing time exceeds 16 hours (24-8 hours). walkbotr
131 then creates a flag to designate data points that are associated with a complete day.

132 After identifying all physical activity bouts and flagging for non-wearing time, the accelerometry
133 data is ready to be merged with GPS data.

134 II. GPS Process

135 Incorporating GPS data to classify bouts as walking or not walking requires four steps. First,
136 walkbotr merges GPS and accelerometry data by timestamp. This merge is complex for two
137 reasons: (1) accelerometry data are typically recorded in local time whereas GPS devices record
138 in UTC time, and (2) whereas accelerometers record activity in consistent epochs from the
139 time they are turned on, GPS devices record when they receive responses from GPS satellites.
140 A fully launched GPS device pings satellites on a regular schedule, often aligned by design with
141 accelerometer epochs. However, when devices re-establish connections with satellites (e.g., after
142 a device restart or after time spent in a tunnel), timestamps may be off-alignment. Accordingly,
143 for each recorded GPS point, walkbotr identifies the accelerometer epoch synchronized local
144 time zone of the GPS points (accounting for daylight savings time), then merges the datasets.
145 When multiple GPS points fall within one accelerometer epoch, walkbotr will assign the
146 latest GPS point within the epoch to reconcile duplications due to temporal alignment.

147 Next, walkbotr calculates a circle to approximate the distance covered by this bout. The
148 circle is centered at the GPS point cloud median and includes the inner 95% of GPS points.
149 An activity bout is considered a dwell bout (i.e., not walking) if, as mentioned above, the
150 individual does not leave a prespecified radius (default setting at 66 feet [Kang-2013]), which
151 may indicate they had not left their home, work, etc. Figure 2 depicts a walk bout and a dwell
152 bout.

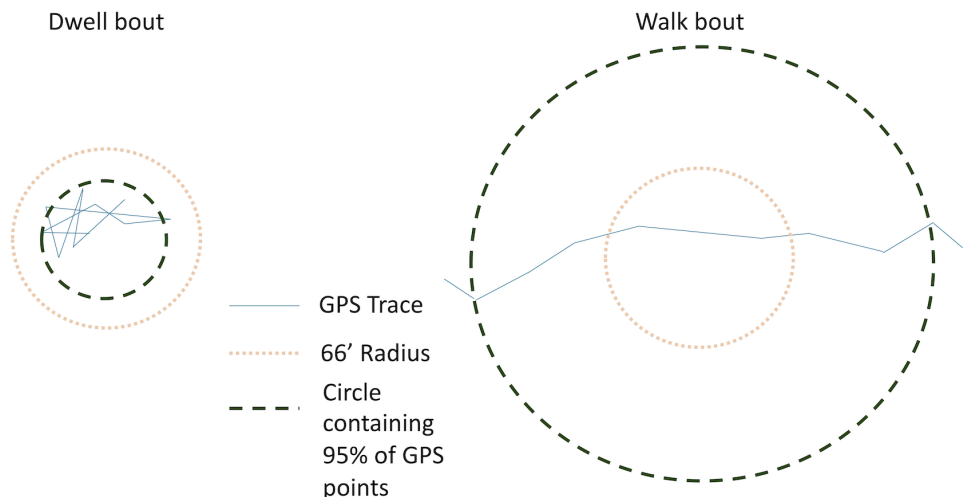


Figure 2: Walk bout (left) and dwell bout (right) show how an individual must leave the dwell bout threshold of 66 feet (shown in pink circle in both plots) in order to be considered a potential walk bout.

GPS data are then evaluated for completeness based on whether they have a sufficient number of GPS records. This is assessed both in terms of the number of GPS observations within a bout as well as the ratio of observations that have GPS data. By default, a bout has sufficient GPS coverage if it has at least five GPS observations and at least 20% of the epochs have a paired GPS observation.

III. Walk Bout Identification

The final step in walkbotr synthesizes the above information and labels each physical activity bout as either a walk bout or a non-walk bout, with a specific label for the different types of non-walk bouts. Each physical activity bout can have just one category, making the order of labeling important. There are six possible categories into which a bout can fall [Kang:2013]. In order, walkbotr applies the following labels:

1. Among all physical activity bouts, bouts without complete GPS data are labeled as a non-walk bout due to incomplete GPS coverage (labeled as **non_walk_incomplete_gps**).
2. Among remaining physical activity bouts, bouts where the median speed exceeds the maximum walking speed are labeled as a non-walk bout due to high speed (labeled as **non_walk_too_fast**). Maximum walking speed defaults to 6 kilometers per hour.
3. Among remaining physical activity bouts, bouts where the median speed falls below the minimum walking speed are labeled as a non-walk bout due to low speed (labeled as **non_walk_too_slow**).
4. Among remaining physical activity bouts, bouts whose GPS data do not exceed a bounding radius of 66 feet are labeled as dwell bouts (labeled as **dwell_bout**).

174 5. Any remaining physical activity bouts are labeled as walk bouts (labeled as **walk_bout**).

175 IV. Outputs

176 From the processed GPS and accelerometry data, a complete, epoch-level dataset (containing
177 epoch time as date-time in the UTC time zone, accelerometry counts per epoch, latitude,
178 longitude, epoch speed, and wearing day complete flag) is used to create two different output
179 datasets:

180 The first output is a full dataset (at the epoch level) – this dataset returns all of the original
181 input data that the user provided, in addition to the new columns that walkbotr created.
182 This dataset is not de-identified, and thus is appropriate for investigators interested in exploring
183 where people walk (e.g., a walkability analysis) [Dalmat:2021].

184 The second output is a summarized dataset (at the bout level), which has been de-identified
185 and collapsed to only include summary walk bout information. This dataset is intended to
186 provide essential walking and physical activity metrics without any identifying information –
187 thus serving as a product that can be easily shared. This dataset can be used in analyses of
188 walking, merged with other key covariates, and used in research studies of external factors
189 that do or do not increase walking in a population (e.g., neighborhood features and their
190 association with walkability can be merged on prior to deidentification and incorporated after
191 identifying features have been removed) [Mooney:2020].

192 The full dataset (at the epoch level) can be seen in Table 1. The summarized dataset (at the
193 bout level) can be seen in Table 2.

Column	Class	Definition
bout	Numeric	This column is a label for each walk bout – each bout is sequentially labeled with a number for easier identification purposes.
bout_category	Character	This column contains the category of the bout, which is described below.
activity_counts	Numeric	Accelerometer counts in counts per epoch (CPE).
bout_start	Date-time	This column contains date-time values in the UTC time zone.
non_wearing	Logical	Boolean flag for whether the user was wearing their device at the time (non_wearing = TRUE).
complete_day	Logical	This is a Boolean column indicating whether the calendar day of data was complete (assessed by determining whether the individual wore their accelerometer for greater than x hours, where x is passed in as a parameter min_wearing_hours_per_day or defaults to 8.)
latitude	Numeric	Latitude coordinate.
longitude	Numeric	Longitude coordinate.
median_speed	Numeric	This column contains the median speed, in km/h, of a given bout.
duration	Numeric	This column contains the length of a bout, in minutes.

194 **Table 1.** Full dataset.

Column	Class	Definition
bout	Numeric	This column is a label for each walk bout – each bout is sequentially labeled with a number for easier identification purposes.

Column	Class	Definition
median_speed	Numeric	This column contains the median speed, in km/h, of a given bout.
bout_category	Character	This column contains the category of the bout, which is described below.
complete_day	Logical	This is a Boolean column indicating whether the calendar day of data was complete (assessed by determining whether the individual wore their accelerometer for greater than x hours, where x is passed in as a parameter. min_wearing_hours_per_day or defaults to 8.)
bout_start	Date-time	This column contains date-time values in the UTC time zone.
duration	Numeric	This column contains the length of a bout, in minutes.

195 **Table 2.** Summarized dataset.

196 The walkbotr package also produces figures describing the walk bouts that are generated,
197 as demonstrated below (Figure 5). This figure shows a walk bout (contained within the gray
198 box) where the accelerometry counts exceed the threshold for being considered active, and an
199 image of the ratio of radii of the bout to the the dwell bout threshold. Given that (1) the
200 activity CPE are consistent with that of walking and (2) the bout area is larger than that of
201 the dwell bout, this is considered a walk bout.

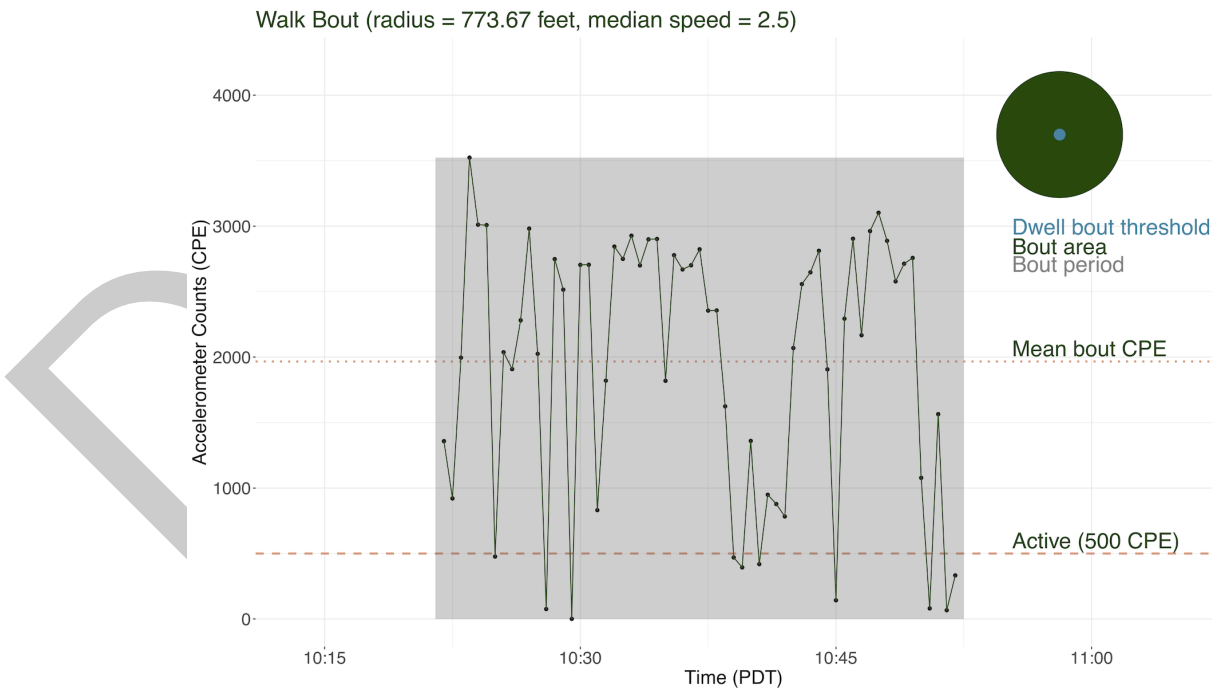


Figure 3: Example of a walk bout.

Conclusions and future directions

walkbotr enables researchers to leverage data collected from monitoring devices such as GPS trackers and accelerometers to more objectively measure walking behavior and its associated impacts on health across diverse geospatial contexts and populations.

By implementing a validated algorithm, walkbotr facilitates not only the identification of walking periods in a consistent and user-friendly manner, thereby bridging the gap between raw data and actionable insights, but also allows researchers to process complex and identifiable monitoring data into shareable insights.

As public health research increasingly embraces data-driven methodologies, the walkbotr package emerges as a tool for researchers to expand their collaborations and datasets. It may serve as an example of how public health researchers can begin to shift towards a new era of collaborative, privacy-conscious, and data-rich health research. By facilitating the extraction of valuable insights from monitoring data without compromising on privacy nor ethics, the package paves the way for a more informed understanding of human behavior, health, and well-being in the world of big data.

Acknowledgements: We thank Amy Youngbloom for her contributions and assistance in vetting these methods.

DMC - Suggested removal

Figure 1 presents a high-leveled dataflow diagram of how walkbotr works. The four modular transformation steps will be described in the following sections as the Accelerometry Process, GPS Process, Walk bout identification, and Summary of walk bouts.

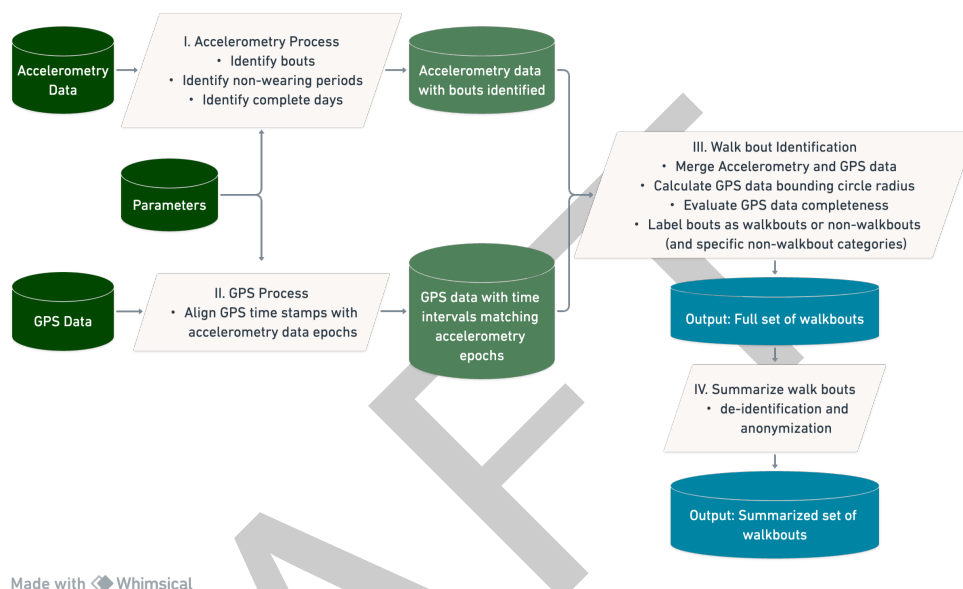


Figure 4: walkbotr process.

Inputs

walkbotr takes in GPS and accelerometry data and processes them to generate walk bouts, using several optional parameters. The following description of these arguments, parameters, and inputs are applicable as of June 2025 and users can check out the walkbotr [CRAN page](#) and [website](#) for any future updates.

Accelerometry data

Accelerometry data are collected via accelerometers and are expected to contain two columns – time and activity counts – where time represents date-time values in the UTC time zone and activity (a numeric column) represents counts per epoch, as described above. walkbotr expects the user to convert time to UTC time for use in the package, but also takes the local time zone as an argument in order to ensure the correct handling of date-time. This format is consistent with the output format from the Actigraph GT3X accelerometer, a device commonly used in research. Refer to Figure 1 for a visualization of accelerometry data.

The accelerometry data are validated and processed. The processed accelerometry data contain the columns described in Table 1.

Column	Class	Definition
activity_counts	Numeric	This column contains the activity counts from the original data, in counts per epoch (CPE).

Column	Class	Definition
time	Date-time	This column contains date-time values in the UTC time zone. (Note: For the determination of the complete_day variable, this column is converted to local time and then back to UTC in order to ensure the correct evaluation of whether a day is complete).
bout	Numeric	This column is a label for each walk bout – each bout is sequentially labeled with a number for easier identification purposes.
inactive	Logical	This is a Boolean column for whether the activity_counts > 500 CPE, which is the threshold for an individual being considered active.
non_wearing	Logical	This is a Boolean column indicating whether the individual was wearing their accelerometer during that period (assessed by determining if the individual was wearing the accelerometer for x consecutive epochs, where x is passed in as parameter non_wearing_min_threshold_epochs or defaults to 40.)
complete_day	Logical	This is a Boolean column indicating whether the calendar day of data was complete (assessed by determining whether the individual wore their accelerometer for greater than x hours, where x is passed in as parameter non_wearing_hours_per_day or defaults to 8.)

Table 1. Processed accelerometry data.

GPS data

Raw GPS data are expected to contain column times, latitude, longitude, and speed – where time represents date-time values in the UTC and speed is in kilometers per hour. These data are consistent with the format used by the QStarz BG-1000XT GPS device commonly used in research [Steel-2021]. The GPS data are essential to determining whether an individual has left a given area and is thus walking. In order to assess this, walkboutr estimates the distance that an individual has traveled and use that to determine if the individual is likely to have left a given area. If an individual meets the activity count requirements to be considered to be on a walk but does not leave a given location, convention is to label this a dwell bout. A dwell bout may be characterized by an individual walking around their home, walking around their work, etc. – the individual is walking, but the individual is not on a walk. This is an important aspect of measuring walking as it differentiates movement in one space from movement through space [Kang-2013].

The processed GPS data consist of a dataset with columns time, longitude, latitude, and speed, where time is now the nearest epoch start time (rather than the precise time stamp of the GPS data point). GPS data are assigned to an epoch start time by rounding down the time associated with the GPS datapoint to the nearest epoch start time. For example, if epochs in the accelerometry data are 30 seconds, the time associated with a GPS data point will be rounded down to the nearest 30-second increment. If there are multiple GPS datapoints within a single accelerometry epoch, the latest GPS data point in that epoch is used. This allows for the integration of the accelerometry and GPS datasets. The columns described here can be found in Table 2.

Column	Class	Definition
time	Date-time	This column contains date-time values in the UTC time zone. (Note: this column is now the nearest time that corresponds with an accelerometry epoch, as described above).
longitude	Numeric	Longitude coordinate.
latitude	Numeric	Latitude coordinate.
speed	Numeric	Current speed in kilometers per hour.

261 **Table 2.** Processed GPS data.

262 Constants and parameters

263 walkboudr contains several optional parameters and constants within the package. The
 264 constants are pre-set in the package, but the parameters can be adjusted if a user wants to. If
 265 a user does not want to adjust any parameters, there is no further specification required.

266 The following parameters are optional, and the default value in the absence of a user specifying
 267 the parameter, is shown in Table 3. These values can be passed as arguments to the top level
 268 functions in the package.

Column	Definition	Default
epoch_length	The duration of an epoch in seconds.	30
active_counts_per_epoch_min	Minimum accelerometer counts for an epoch to be considered active (vs. inactive).	500
minimum_bout_length	Minimum number of epochs for a period of activity to be considered as a potential bout.	10
local_time_zone	Local time zone of the data - data come in and are returned in UTC, but local time zone is used to compute complete days.	America/Los_Angeles
maximum_number_consec_inactive_epochs_in_bout	Number of consecutive epochs that can be labeled as inactive during a bout without ending the bout.	3

269 **Table 3.** Parameters.

270 The constants can be found in Table 4.

Column	Definition	Default
non_wearing_min_threshold_epochs	Number of consecutive epochs with activity counts of 0 that constitutes a period where the device is not worn	40
min_wearing_hours_per_day	Minimum number of hours in a day an individual must wear an accelerometer for the day to be considered complete.	8
min_gps_obs_within_bout	Minimum number of GPS observations within a bout for that bout to be considered to have complete GPS data.	5
min_gps_coverage_ratio	Minimum ratio of data points with GPS data vs. without GPS data for the bout to be considered to have complete GPS data.	0.2

Column	Definition	Default
dwellobout_radii_quantile	Threshold for outlying GPS data points - any data points above the 95th percentile are considered outliers.	0.95
max_dwellobout_radii_ft	Maximum radius, in feet, of a bounding circle that would be considered a dwell bout (rather than a potential walk bout).	66
min_dwellobout_obs	Minimum number of observations to consider something a potential dwell bout.	10
max_walking_cpe	Maximum CPE value before the accelerometer is considered to be picking up on an activity more intense than walking.	2863
min_walking_speed_km_h	Minimum speed considered walking.	2
max_walking_speed_km_h	Maximum speed considered walking.	6

271 **Table 4.** Constants.

272 Figure 4 is a high-level summary of this process:

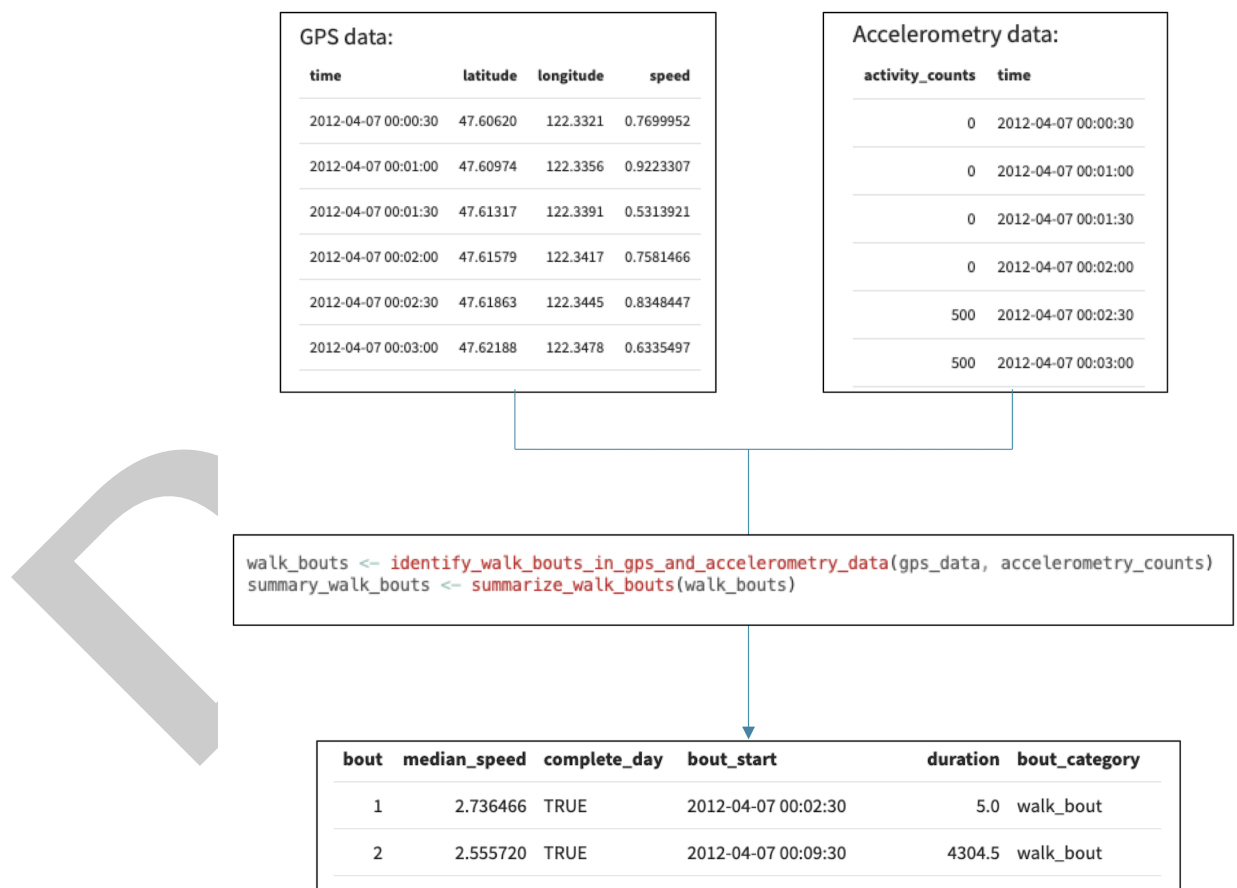


Figure 5: walkbouts process using simulated data included in package.

273 Discussion

274 Contribution of walkboudr to public health research

275 walkboudr allows researchers to evaluate their personal monitoring data using a consistent and
276 precise definition of a walk bout, allowing for research efforts to be focused on the implications
277 of walking. This implementation of the algorithm will aid physical activity researchers in
278 evaluating changes in walking behavior in their populations of interest, thereby facilitating
279 research in this space.

280 An example of walkboudr use: TRAC study

281 Our most recent application of this package is using the TRAC collected data beginning in
282 2008 with the goal of assessing the ways in which new light-rail changes in King County
283 affected physical activity, particularly walking. Participants in the TRAC study wore a uni-axial
284 or tri-axial accelerometer (GT1M or GT3X Actigraph LLC, Fort Walton Beach, FL) and carried
285 a GPS device (DG-100 GPS data logger, GlobalSat, Taipei, Taiwan or QStarz BT1000-XT)
286 [Kang:2018]. The data from the TRAC study were cleaned and processed to contain the
287 columns required by walkboudr and were then processed to identify walk bouts in the data.
288 These walk bouts were then used, for example, to identify whether people walk more near
289 new transit stops [Huang:2017], determine which residential neighborhood features were most
290 correlated with walking [Mooney:2020], and compare walkability metrics to determine which
291 best predicted walking [Dalmat:2021].

292 However, the code for walk bout identification for these previous papers was developed only
293 for the TRAC study. walkboudr implements the bout identification algorithm in a re-usable
294 way and deidentifies the data; it allows all researchers to use the same algorithm and share
295 common data, and provides efficient and consistent processing of data to estimate walking.

296 Contribution of walkboudr to data privacy and ethics practices

297 In addition to aiding physical activity researchers in their ability to evaluate walking and thus
298 outcomes with which exposure can be tied, walkboudr represents an example of an R package
299 in the public health research space that allows researchers who work with personally identifying
300 and sensitive data to collaborate with other researchers safely and without risk of privacy
301 breaches.

302 walkboudr does this by collapsing all identifying information within its processing and outputting
303 a dataset that contains all relevant walking information without any personal identifiers, exact
304 GPS coordinates, nor exact accelerometer values – making it impossible to trace any bouts
305 back to individuals. Given the sensitive nature of GPS and accelerometry data, walkboudr
306 thus offers anonymous set of walk bouts upon which researchers may collaborate. For example,
307 it allows researchers looking at walking in King County to collaborate with researchers looking
308 at walking in other counties, as the light-rail continues to expand, in the case of the TRAC
309 study, without any risk of privacy breaches. The ability to de-identify, anonymize, and label
310 GPS data opens many more avenues to collaboration in this space.

311 References